

A Unique Encoding of Satisfying Assignments for Balanced CNFs

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ABSTRACT

In this paper we introduce and prove the SAT Equation Theorem, which states that every balanced CNF formula over n Boolean variables can be uniquely characterized by the number

$$S = \sum_{C \in \mathcal{F}} 2^{\left(\sum_{x \in C, x > 0} 2^{n-|x|}\right)},$$

where $\mathcal{F}$ is a CNF formula whose every clause is balanced (i.e., it contains exactly one literal per variable) and the inner sum encodes the sign pattern of C . In our encoding convention, a bit value 1 in the binary expansion of S indicates an assignment that does not satisfy $\mathcal{F}$ (unsat), while a 0 indicates satisfaction. The theorem asserts that when written with at least 2^n bits (so that every Boolean vector corresponds to a unique position), S encodes exactly the truth table of $\mathcal{F}$. We give here a detailed statement and proof, along with remarks on its significance.

CCS CONCEPTS

• Theory of computation $\rightarrow$ Logic;

KEYWORDS

Satisfiability; Number Theory

1 INTRODUCTION

In propositional logic, formulas written in Conjunctive Normal Form (CNF) play an essential role. A particular class of CNF formulas—called *balanced* formulas—is one in which every clause contains exactly one literal per variable. In a balanced clause the n literals are chosen as $\pm 1, \pm 2, \dots, \pm n$ (with no repetition) so that each variable appears once with either a positive or negative sign.

In this work we show that for any such balanced CNF formula $\mathcal{F}$ over n variables, one can define the number

$$S = \sum_{C \in \mathcal{F}} 2^{\left(\sum_{x \in C, x > 0} 2^{n-|x|}\right)},$$

which has the remarkable property that its binary expansion (written with at least 2^n bits) exactly encodes the truth table of $\mathcal{F}$ by marking those assignments which do not satisfy $\mathcal{F}$ (with a 1) and leaving zeros in all other positions. In other words, S uniquely determines all satisfying assignments.

This paper presents a clear statement and rigorous proof of Riveros' Theorem along with some discussion of its significance and potential applications.

2 PRELIMINARIES

We assume the reader is familiar with basic concepts in propositional logic. A Boolean variable x_i (for $i = 1, \dots, n$) takes a value in $\{0, 1\}$, where we adopt the convention that 0 represents True and 1 represents False when evaluating formulas.

A *literal* is either a positive occurrence ($+i$) or a negative occurrence ($-i$) of variable x_i . A clause is a disjunction (OR) of literals; a CNF formula $\mathcal{F}$ is a conjunction (AND) of such clauses. In a balanced CNF, every clause contains exactly one literal per variable. Consequently, each clause $C \in \mathcal{F}$ can be uniquely represented by its sign vector:

$$v(C) = (v_1, v_2, \dots, v_n), \quad \text{where } v_i = \begin{cases} 1 & \text{if } +i \in C, \\ 0 & \text{if } -i \in C. \end{cases}$$

It immediately follows that the number

$$T(C) = \sum_{i=1}^n v_i 2^{n-i}$$

is just the binary representation of $v(C)$ (with the most significant bit corresponding to variable 1).

Furthermore, under standard evaluation rules for disjunctions [?], a clause is false if and only if every literal is false. For balanced clauses this happens in exactly one case: if we assign

$$a_C(i) = \begin{cases} 1 & \text{if } v_i = 1, \\ 0 & \text{if } v_i = 0, \end{cases}$$

then every literal in C evaluates to false and hence C is unsatisfied. Notice that the binary number corresponding to this unique falsifying assignment is exactly $T(C)$.

3 THEOREM STATEMENT

THEOREM 3.1. *Let $\mathcal{F}$ be a CNF formula over n Boolean variables in which every clause is balanced (i.e., each clause contains exactly one literal per variable). Define*

$$S = \sum_{C \in \mathcal{F}} 2^{\left(\sum_{x \in C, x > 0} 2^{n-|x|}\right)}.$$

If we write S in binary using at least 2^n digits (so that every assignment in $\{0, 1\}^n$ corresponds to a unique bit position), then the resulting number encodes the truth table of $\mathcal{F}$ with respect to satisfying assignments: a 1 indicates an assignment that does not satisfy $\mathcal{F}$ and a 0 indicates satisfaction. In particular, S uniquely determines all satisfying assignments.

4 PROOF

The proof proceeds in three main steps.

Step 1: Encoding Each Clause

Since every clause $C \in \mathcal{F}$ is balanced, there exists a unique sign vector

$$v(C) = (v_1, v_2, \dots, v_n), \quad v_i = \begin{cases} 1 & \text{if } +i \in C, \\ 0 & \text{if } -i \in C, \end{cases}$$

and hence

$$T(C) = \sum_{i=1}^n v_i 2^{n-i}.$$

This $T(C)$ is simply the binary encoding of the sign pattern for clause C .

Step 2: The Unique Falsifying Assignment

Under standard evaluation rules, a clause (being a disjunction) is false if and only if every literal is false. For balanced clauses this occurs uniquely when each variable is assigned as follows:

$$a_C(i) = \begin{cases} 1 & \text{if } v_i = 1, \quad (\text{since } +i \text{ becomes false}), \\ 0 & \text{if } v_i = 0, \quad (\text{since } -i \text{ becomes false}). \end{cases}$$

Thus the unique assignment a_C that falsifies clause C corresponds exactly to the binary number $T(C)$.

Step 3: Forming and Interpreting S

By definition,

$$S = \sum_{C \in \mathcal{F}} 2^{T(C)}.$$

Assume that $\mathcal{F}$ contains no repeated clauses (if there are duplicates, one may adjust the argument to account for carries in binary addition). Then each clause contributes a distinct power of two. Writing S in base-2 with at least 2^n digits, the only positions where a 1 occurs are those corresponding to some $T(C)$. That is, an assignment $a \in \{0, 1\}^n$ appears as a 1 (i.e. unsat) if and only if there exists a clause C with $T(C) = a$.

Since $\mathcal{F}$ is satisfied by an assignment precisely when none of its clauses are falsified (i.e. when the corresponding bit in the binary expansion is 0), it follows that the binary representation of S encodes exactly the truth table of $\mathcal{F}$ —with 1's marking unsatisfying assignments and 0's marking satisfying ones. Thus, from S one may uniquely recover all satisfying assignments.

5 CONCLUSION

We have shown that for any balanced CNF formula over n Boolean variables the number

$$S = \sum_{C \in \mathcal{F}} 2^{\left(\sum_{x \in C, x > 0} 2^{n-|x|}\right)}$$

encodes the truth table of $\mathcal{F}$ in a unique way. In our encoding convention, each bit corresponds to an assignment (with 1 meaning unsatisfied and 0 meaning satisfied), so that S completely characterizes the set of satisfying assignments. This result not only offers a compact representation for balanced CNF formulas but also provides new insights into their structure, with potential applications in automated reasoning and logic synthesis.